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REVIEW PAPER

Advancing inclusive on-farm testing approaches to facilitate maize breakthrough product development in sub-Saharan Africa

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Abstract

Developing breakthrough products that deliver a minimum 25% productivity improvement in farmers' fields requires that breeding pipelines more closely align with farmer and end-user needs and the target population of environments (TPEs). In the case of maize (*Zea mays*), climate change is accelerating climatic shifts within the TPE, potentially limiting future realized genetic gain. On-farm breeding approaches can play a valuable role in supporting breeding programs transitioning toward breakthrough products. Increased sampling of the TPE by moving selection into farmers' fields can overcome the challenges of climatic changes within the TPE and extensive farmer diversity, whilst providing information to refine managed environment trials designed to sample the TPE. This review highlights advances in inclusive host farmer sampling, sparse testing strategies and phenotyping to facilitate scaling of on-farm testing within breeding pipelines.

Keywords: Breakthrough products, breeding, managed environment trials, on-farm testing, participatory plant breeding, participatory varietal selection, social inclusion, target population of environments.

Abbreviations: CGIAR, Consultative Group for International Agricultural Research; E, environment; FAIB, farm-as-incomplete-block; G, genotype; GRM, genomic relationship matrix; G×E×M, Genotype×Environment×Management; M, management; *N*, nitrogen; NARES, National Agricultural Research and Extension Systems; PPB, participatory plant breeding; PVS, participatory varietal selection; SSA, sub-Saharan Africa; TPE, target population of environments.

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Introduction

The move by the Consultative Group for International Agricultural Research (CGIAR) and National Agricultural Research and Extension Systems (NARES) toward breakthrough products sets a new direction for public breeding programs. New crop varieties must deliver measurable impact on national production, requiring at least 25% improvement in productivity and/or value in farmers' fields and be 'adoption ready' with the potential to be rapidly deployed at scale (Coaldrake and Werner, 2025). Demonstrated productivity (or profitability) in farmers' fields will be critical for the release of breakthrough products. Meeting these targets will require breeding pipelines more closely aligned with farmer and end-user needs (Donovan *et al.*, 2025) and shifting environments within the target populations of environments (TPE) (Cooper *et al.*, 2021). On-farm testing offers a major opportunity to support the development and release of breakthrough products by evaluating candidate varieties directly within farmers' production environments. Advances in representative host farmer sampling tools, quantitative genetics to support on-farm sparse testing strategies, and high-throughput phenotyping can help scale on-farm testing within breeding pipelines. In maize (*Zea mays L.*), this creates a greater opportunity to develop new varieties for drought-prone areas within increasingly complex target environments.

Drought is a major driver of yield loss and inter-annual yield variability in maize production across sub-Saharan Africa (SSA) (Ray *et al.*, 2015; Tall *et al.*, 2023; Marcos-Garcia *et al.*, 2024). About 40% of SSA's maize area is exposed to occasional drought stress, causing yield losses of 10% to 25%, while around a quarter of the maize area faces frequent drought stress (Tesfaye *et al.*, 2016). In this region, maize is primarily rainfed and central to food security and livelihoods (Mulungu *et al.*, 2025). For smallholder farming households, yield loss due to drought negatively affects household food security, economic well-being, and resilience, while contributing to risk adversity (Mulungu *et al.*, 2025), poorer mental health outcomes (Ae-Ngibise *et al.*, 2021; Burrows *et al.*, 2024), and increased vulnerability (Kamali *et al.*, 2018). Drought tolerance is therefore a key trait in most maize target product profiles in SSA.

The TPE can be defined as the set of all farms and future seasons in which the varieties produced by a breeding program will be grown (Comstock, 1977; Chenu *et al.*, 2011; Werner *et al.*, 2025) and, therefore, is the environment in which realized genetic gain translates into improved production and livelihoods (van Eeuwijk *et al.*, 2001). Within the TPE, the crop production environment can be partitioned into a largely non-controllable biophysical component, i.e. the environment (E), defined by factors such as weather and soil characteristics, and a controllable management component (M), characterized by a farmers' agronomic decisions and practices, which are shaped by socio-economic factors such as production orientation, aspirations, and wealth. This distinction provides the basis for dissecting genotype-by-environment (G×E) interactions into genotype-by-environment-by-management (G×E×M) interactions, enabling breeding programs to leverage beneficial genotype–management interactions to improve selection and variety deployment. (Ceccarelli, 1989, 1994; Chenu *et al.*, 2011; Cooper *et al.*, 2021, 2022, 2023). Management practices used by farmers are often dynamic, varying by season (Hamadziripi *et al.*, 2025) and market prices (Technow *et al.*, 2026). The delineation of a TPE is a dynamic, iterative process requiring revision as new environmental information emerges and breeding objectives advance (Cooper and Messina, 2023). Maize TPEs in Eastern and Southern Africa were initially delineated using day length, maximum temperature, maturity, preferred grain color and texture, and production constraints, including drought, low nitrogen, diseases, and insects (CIMMYT, 1989). As Geographic Information System (GIS)-derived data on day length, mean temperature, and precipitation over a four-month growing season became available, TPEs were revised to incorporate these variables (Hartkamp *et al.*, 2000). In Southern Africa, TPEs were further refined using additional environmental characteristics (subsoil pH and estimated nitrogen application) and climatic data (total rainfall and maximum temperature over a five-month period) together with sequential retrospective pattern analysis of multi-environment trial data (Setimela *et al.*, 2005; Bänziger *et al.*, 2006).

CIMMYT maize's early TPE delineation laid the foundation for designing test environments that reflected key components of the production conditions targeted by the breeding program. Direct selection for grain yield under managed drought stress trials represented a major advance in breeding for drought-prone maize environments in Eastern and Southern Africa (Bänziger and Diallo, 2001). Until the mid-1990s, maize breeding primarily focused on optimal conditions, despite the prevalence of drought and low soil fertility in farmers' fields (Bänziger and Diallo, 2001). Recognizing the need to develop varieties that better met the real-world needs of smallholder farmers, CIMMYT and partners initiated a large-scale breeding program in 1997 for drought-prone, low input, mid-altitude environments in Southern Africa (Bänziger and Diallo, 2001; Bänziger *et al.*, 2006). This strategy was based on earlier research that showed direct selection for grain yield under managed flowering-stage drought stress resulted in higher gains than selection under optimal conditions or randomly occurring drought stress (Bolaños and Edmeades, 1993; Byrne *et al.*, 1995). Flowering-stage stress was targeted as the most sensitive stage to drought stress in maize (Edmeades *et al.*, 1989) and kernel number per plant, a key driver of grain yield, is determined at this stage (Fischer and Palmer, 1984). Managed drought stress screening was conducted in the off-season to allow drought stress to be imposed, with irrigation stopped approximately 50 days before anthesis. Target yields

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under managed stress were initially 1 to 3 t ha^{-1} (Bänziger *et al.*, 2006) and later increased to 2 to 4 t ha^{-1} (Prasanna *et al.*, 2022). This approach produced hybrids that outperformed commercial checks within the TPE (Bänziger *et al.*, 2006; Setimela *et al.*, 2017), including under extreme El Niño-induced drought stress (Setimela *et al.*, 2018), and resulted in genetic gain in yield under managed drought stress (Masuka *et al.*, 2017; Prasanna *et al.*, 2022; Tarekegne *et al.*, 2023).

The impact of CIMMYT and partner's breeding efforts to increase drought tolerance is reflected in the commercialization of more than 450 drought tolerant maize varieties in SSA (Chivasa *et al.*, 2022), with approximately 8 million ha planted and 9.7 million households benefiting (Cairns and Prasanna, 2018; Chivasa *et al.*, 2022; Boddupalli, 2024). Genetic gain under managed drought stress was estimated at 32.5 kg ha^{-1} yr^{-1} between 2000 and 2010 (Masuka *et al.*, 2017). A subsequent study of genetic trends from 2013 to 2020 estimated gains under managed drought stress of 45, 64, and 54 kg ha^{-1} yr^{-1} in the early maturity pipeline in Southern Africa, the early maturity pipeline in East Africa, and the intermediate maturity pipeline in East Africa, respectively (Prasanna *et al.*, 2022). Beyond yield gains, a growing body of literature connects the cultivation of stress tolerant maize varieties to a range of positive livelihood indicators, including increased farm production (Amondo *et al.*, 2019; Gebre *et al.*, 2025), strengthened livelihoods (Lunduka *et al.*, 2019; Gebre *et al.*, 2021; Katengeza and Holden, 2021), lower household food insecurity (Euler *et al.*, 2026), improved child nutrition outcomes (Zeng *et al.*, 2017), and reduction in depth of poverty at the household level (Amondo *et al.*, 2019; Ngoma *et al.*, 2025). Notably, more resource-constrained households have been shown to gain higher benefits from drought tolerant maize (Gebre *et al.*, 2021). Furthermore, beyond farm-level gains, improved varieties have also been shown to contribute toward broader environmental benefits, including reduced cropland expansion, lower biodiversity loss, and decreased terrestrial carbon emissions (Baldos *et al.*, 2025).

To sustain historic rates of genetic gain, breeding programs must keep pace with rapidly changing environmental conditions. Climate change is accelerating climatic shifts within current TPEs and increasing the complexity of stress combinations that maize varieties will be exposed to. For maize breeding in SSA, patterns of drought are changing in terms of timing, duration, and co-occurrence with other stresses (Lesk *et al.*, 2022). Shifts in dry-wet spell patterns, rising temperatures, and increased frequency of dry spells (Thoithi *et al.*, 2021; Marcos-Garcia *et al.*, 2024) are changing the production environments in which new varieties must deliver value. Compounding this challenge, the impact of drought on maize yields is exacerbated when it occurs alongside heat stress (Lobell *et al.*, 2011; Cairns *et al.*, 2013), and the co-occurrence of drought and heat stress has become more frequent under climate change and El Niño–Southern Oscillation (ENSO)-related extreme events (Lesk *et al.*, 2022; Cai *et al.*, 2025). Climate projections suggest a two-fold increase in the probability of the co-occurrence of drought and heat stress in maize environments in SSA, with the largest increases in hot-dry conditions expected in Southern Africa (Heino *et al.*, 2023). Multi-factorial abiotic and biotic stress combinations in model species are known to be unique and cannot be extrapolated from tolerance to individual stresses (Zandalinas and Mittler, 2022). Shifts in the frequency of co-occurrence of climate-related stresses and environment-types of the TPE under climate change have implications for managed stress screening protocols in terms of effectively sampling the TPE. While managed drought and low nitrogen stress screening remain essential for generating genetic gain under key production constraints, developing breakthrough products will require breeding programs to refine selection environments in line with the rapidly changing conditions within the TPE.

Beyond changes in climate and stress patterns, the extensive farm diversity and resulting G×E×M interactions within the TPE remain challenging to represent adequately through managed selection environments. Smallholder farms are highly diverse in terms of production orientation, aspirations, and management practices (Balogun *et al.*, 2022; Thuijsman *et al.*, 2022). For example, a survey of 2000 smallholder maize farmers in Zimbabwe revealed significant diversity in agronomic practices (Cairns *et al.*, 2026). In SSA, the maize yield gap (the difference between actual yields on farmers' fields relative to the potential yield) is estimated to be over 80% (Aramburu-Merlos *et al.*, 2024). Sub-optimal fertilizer use is the largest and most persistent driver of the maize yield gap in this region (Leonardo *et al.*, 2026). The average fertilizer application within maize plots was 45.4 kg ha^{-1}, however there was large variation in nitrogen (N) management practices between farms. The average number of times maize plots were weeded was 2.12, however it ranged from no weeding to up to five times (Fig. 1B). While mono-cropping of maize was prevalent, specific groups of smallholder farmers grew maize in association with other crops (Fig. 1C). Forty-one percent of women-managed maize plots were intercropped (Fig. 1D), compared to 37% of jointly managed maize plots (Fig. 1E) and only 25% of male-managed maize plots (Fig. 1F). This diversity in management practices is likely to translate into large soil fertility gradients, production environments, and, ultimately, yields (Masvaya *et al.*, 2010; Chikowo *et al.*, 2014). Given this complexity, it is increasingly challenging for testing strategies to adequately sample the TPE, with unintended misalignment of selection environments within the TPE likely to reduce realized genetic gain within the target environment (Atlin *et al.*, 2001; Werner *et al.*, 2025).

Achieving breakthrough products with at least 25% improvement in productivity and/or value requires breeding programs to move beyond incremental improvements and adopt approaches capable of delivering substantially larger gains in farmers' fields. Experiences from commercial maize breeding illustrate how breeding strategies can be redesigned to achieve

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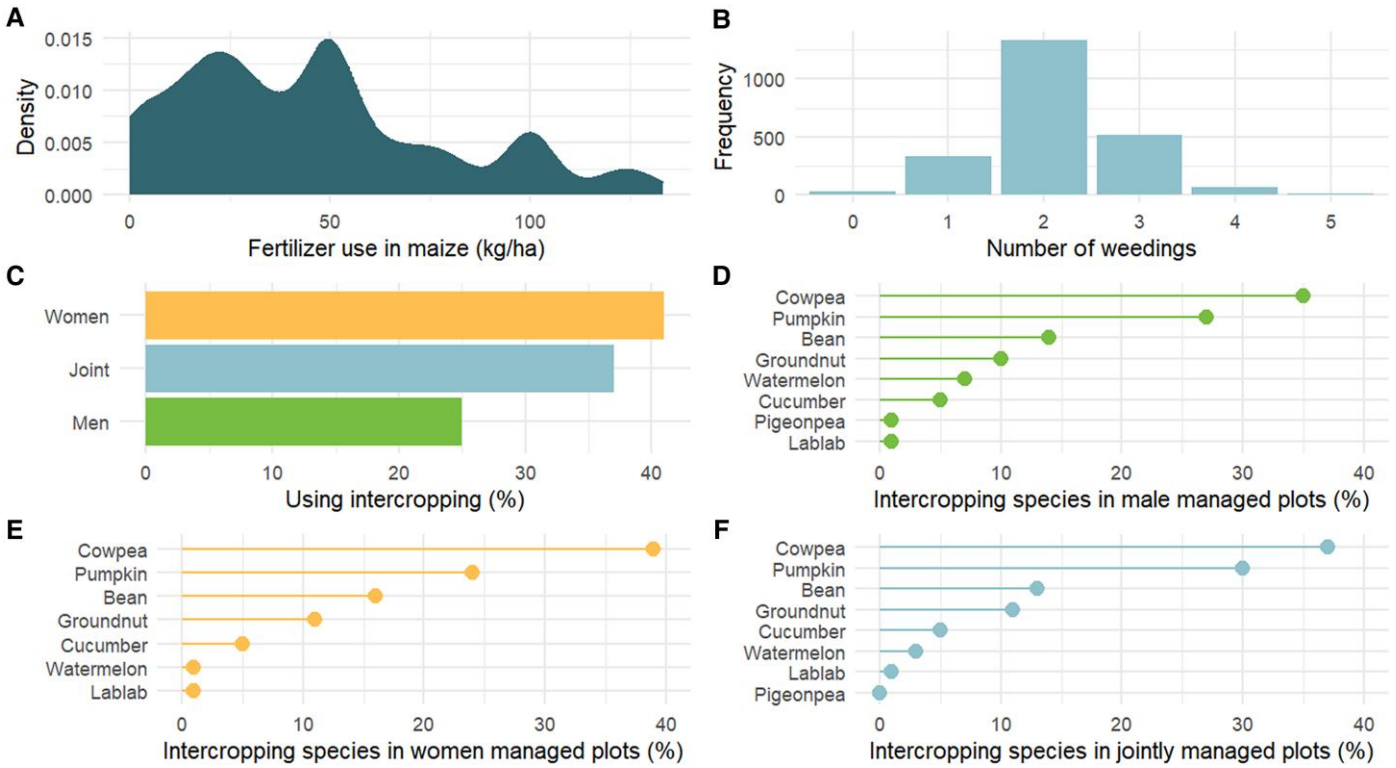


Fig. 1. Agronomic management practices in *Zea mays* (maize) plots derived from a survey of almost 2000 smallholder farmers in Zimbabwe including (A) average fertilizer use, (B) total number of times maize plots were weeded, (C) percentage of plot managers practicing intercropping within maize plots, (D) intercropping species used in maize production within male-managed plots, (E) intercropping species used in maize production within women-managed plots, and (F) intercropping species used in maize production within jointly-managed plots. The study was conducted in September 2021 in five provinces (Mashonaland West, Midlands, Mashonaland East, Manicaland and Masvingo). Covering agroecological zones 2 (700–1050 mm per annum), 3 (500–800 mm per annum) and 4 (450–650 mm per annum (Moyo, 2000). Cowpea, *Vigna unguiculata* L.; Pumpkin, *Cucurbita* spp.; Bean, *Phaselous vulgaris;* Groundnut, Arachis hypogaea; Cucumber, *Cucumis metuliferus*; Watermelon, Citrullus lanatus; Pigeon pea, *Cajanus cajan* L.; Lablab, *Lablab purpureus*.

this goal. AQUAMax drought-tolerant maize varieties released in the United States provide a concrete example of breakthrough products. Key components in their development included the delineation of environment-types within TPE, careful alignment of the managed selection environments within the TPE (Cooper *et al.*, 2014), and extensive on-farm testing (Gaffney *et al.*, 2015). Multi-trait phenotypic data from trials in managed environments were used to continuously update genomic prediction models, enabling rapid-cycle development and testing of inbred lines and hybrids. For crop breeding programs in SSA, moving selection into farmers' fields could help overcome the challenges of capturing the rapidly shifting environment-types and extensive farmer and agronomic management diversity within the TPE (Witcombe *et al.*, 2005). By moving selection closer to the biophysical environments, agronomic management practices, and farmer priorities that define variety success, on-farm testing approaches can provide a direct way to connect breeding pipelines with the dynamic TPE, while concurrently refining the environment composition of the TPE and, subsequently, improving the design of managed environmental trials to represent the TPE.

In public plant breeding programs, on-farm testing approaches offer a major opportunity to support the move to breakthrough products by evaluating candidate varieties directly within farmers' production environments. On-farm testing approaches include participatory plant breeding (PPB), which involves farmers throughout the breeding pipeline, and participatory varietal selection (PVS), which involves farmers in late-stage testing (Witcombe *et al.*, 1996). On-farm testing approaches vary in the level of involvement of farmers, from farmers' selection of candidates for advancement to farmers only providing varietal and trait preferences (Ceccarelli and Grando, 2020). In this review we focus on on-farm testing to evaluate candidates within farmers' fields and soliciting farmers feedback on varietal and trait preferences.

The value of late-stage on-farm testing is well established within the commercial sector. On-farm variety testing is routinely used in highly competitive market segments by the commercial seed industry to assess candidate variety performance under real-world conditions, accelerate product design feedback loops, generate pre-commercial insights, and provide head-to-head comparisons with competing varieties within

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the same market segment (Gaffney *et al.*, 2015; Pires *et al.*, 2024;). While participatory breeding approaches have a long history within the public sector (Sperling *et al.*, 1993, 2001; Witcombe *et al.*, 1996; Witcombe, 2000; Ceccarelli and Grando, 2020; de Sousa *et al.*, 2024), in general, on-farm testing has been used less systematically within breeding programs, often on a project-specific basis (Collinson *et al.*, 2022; Hamadziripi *et al.*, 2025). Notable exceptions include participatory breeding approaches for sorghum (*Sorghum bicolor L.*) in West Africa (Isaacs *et al.*, 2024), barley (*Hordeum vulgare L.*) in Syria and North Africa (Ceccarelli *et al.*, 2001) and rice (*Oryza sativa L.)* in South Asia (Witcombe *et al.*, 1999). Furthermore, information on farmer preferences generated through on-farm testing has been used primarily on an *ad hoc* basis to refine product design (Rice *et al.*, 2024).

Despite extensive evidence showing new varieties developed by incorporating on-farm testing into development meet the needs of farmers and are widely adopted (Witcombe, 2000; vom Brocke *et al.*, 2020), there has been limited systematic integration into breeding pipelines (Witcombe and Yadavendra, 2014). Path dependency, whereby technology development 'may be quite flexible when it first develops, but over time steadily more fixed pathways become established' (McGuire, 2008; Conti *et al.*, 2021), and hesitancy of breeding teams to change established breeding schemes (Witcombe and Yadavendra, 2014) have been cited as key constraints toward the mainstreaming of on-farm testing within public breeding programs. The cost of running a large-scale on-farm network has also been a key constraint to mainstreaming on-farm testing within breeding pipelines (Bänziger and de Meyer, 2002; de Sousa *et al.*, 2024).

The move from varieties that offer incremental increases in performance to breakthrough products, combined with donor-driven mandates to institutionalize late-stage on-farm testing to support varietal replacement and the formation of an overarching body (CGIAR Breeding for Tomorrow) to standardize methodologies across CGIAR-NARES breeding networks, provides a strong rationale for integrating on-farm testing into breeding pipelines in SSA.

In recognition of the current transition of crop breeding programs in SSA toward breakthrough products, this review highlights advances in representative farmer sampling and quantitative genetics to enhance the effectiveness and scalability of on-farm testing within breeding pipelines. High-throughput phenotyping approaches that can be cost-effectively scaled within on-farm testing networks remains a bottleneck in on-farm testing, and key requirements are highlighted.

Sampling strategies to ensure host farmers are representative of the target population of environments

Central to the validity of on-farm testing strategies is how representative host farmers are of the wider TPE (Witcombe *et al.*, 2005). Ensuring adequate farmer representation requires sampling across both the range of biophysical conditions and agronomic management practices that characterize the TPE, as differences in management can create substantial variation in the environments experienced by crops even under similar biophysical conditions. This is particularly important for drought, where management practices such as sowing date and planting density can alter both the developmental stage at which crops are exposed to water stress and the rate at which soil water is depleted (No single profile defines farmers (Acevedo *et al.*, 2020) and, when host farmers are not representative of the wider population they are intended to represent, varieties that delivered measurable impact in on-farm validation trials may deliver less benefit under the real-life conditions of the diverse range of farmers within the TPE. This will ultimately result in low adoption, with new varieties that appear promising in trials failing to translate into widespread uptake.

Smallholder farmers are highly diverse in terms of wealth, production orientation, livelihood strategies, and constraints, even within a single community (Dowd-Uribe *et al.*, 2024; Schnurr *et al.*, 2025; Cairns *et al.*, 2026). Given this diversity, in socio-economic research, sampling strategies and reporting of farmer characteristics are considered fundamental. However, within crop breeding literature, while environmental characteristics of experimental research stations or on-farm trial networks are routinely documented, limited or no information is provided on the sampling methods used to select host farmers for on-farm trials (Cairns *et al.*, 2026) or the subsequent demographic characteristics (e.g. sex, age, level of education, wealth and assets, and market orientation) of host farmers (Toffolini and Jeufroy, 2022) and their agronomic management practices. As a result of these reporting gaps, it is often difficult to determine whether on-farm testing networks accurately represent farmers within the TPE, or adequately account for farmers' diverse aspirations, needs, and preferences. Furthermore, this reduces the external validity of results (Kool *et al.*, 2020). Studies have shown that host farmers in agronomic research tend to be more prosperous and better educated than the broader underlying population (de Roo *et al.*, 2019; Ochieng *et al.*, 2019; Laajaj *et al.*, 2020; Hunt *et al.*, 2025). This is, in part, associated with researchers' desire to validate technologies within the TPE (Laajaj *et al.*, 2020) and well-off farmers are often easier to access (de Roo *et al.*, 2019) but can ultimately result in less promising results upon scaling. To address this, training of key actors responsible for host farmer recruitment on the objectives of on-farm testing for crop improvement strategies and the rationale for broad representation of different farmer types is critical to reduce recruitment bias (Sharma *et al.*, 2025; Cairns *et al.*, 2026).

In host-farmer sampling methods, the handful of studies that report their methodology reveal that nonrandom selection is the dominant approach, particularly for plot selection (Laajaj *et al.*, 2020). Non-probability sampling, such as nomination by technicians, agricultural extension agents, community

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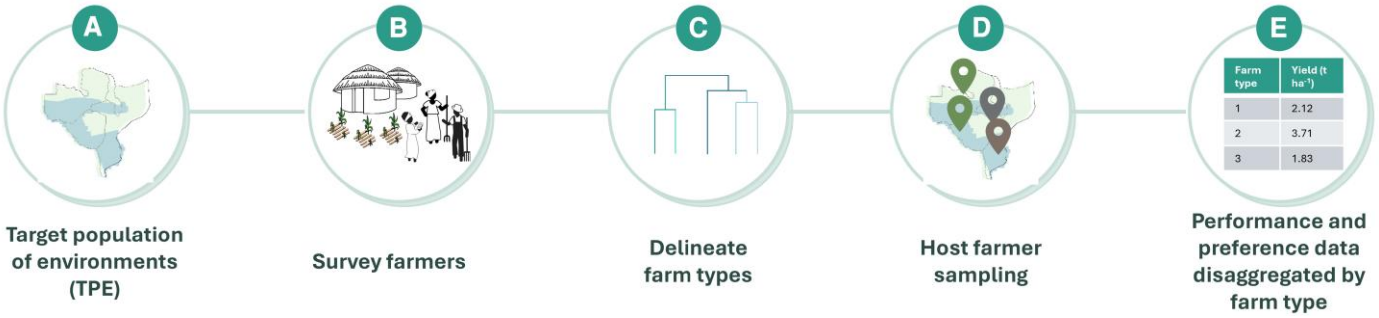


Fig. 2. A stratified sampling approach can be used to ensure host farms represent the wider population they are selected to represent farmer preferences by farm type can also be included in product advancement meetings and publications to facilitate replication of results (Kool *et al.*, 2020). To identify key farm types, the following steps are undertaken. (A) The delineation of the target population of environment (TPE) provides the initial information to guide sampling of host farmers. On-farm trials should adequately cover the TPE to ensure wide sampling. (B) Within each identified region, farmers are randomly sampled from village lists and surveyed about variables associated with inconsistency in farming systems and crop production. (C) Survey data is used to construct a typology and farm types delineated. (D) A stratified sampling approach using region and farm type to select representative host farmers, with selection based on the proportion of farm types or oversampling to increase the representation of specific types of farmers. (E) Host farmer characteristics should be included in analysis and reporting to provide greater understanding of market segments and facilitate replication of results (Kool *et al.*, 2020).

members, or farmer groups, can unintentionally introduce bias, limiting the representativeness of experimental outcomes (Freeman, 2001; Kool *et al.*, 2020). A key consequence is that certain groups, such as women, tend to be underrepresented (van Etten *et al.*, 2019), while others, including wealthy and powerful individuals, are overrepresented (Hailemichael and Haug, 2020; Ochieng *et al.*, 2022).

Recruitment bias can be reduced by setting explicit targets for participation of different groups of farmers. This has proved an effective approach for ensuring adequate representation of women as host farmers (Nanyonjo *et al.*, 2024; Hamadziripi *et al.*, 2025; Kang *et al.*, 2025; Voss *et al.*, 2025; Cairns *et al.*, 2026). However, deliberately targeting female plot managers has the potential to create unintended pressures on the women involved by overlooking household power dynamics. The use of an intrahousehold approach (Doss, 2013), whereby spouses or other key household decision-makers are involved in the trial hosting process to some degree, may help foster household cooperation and reduce potential tensions that may arise when women plot managers are specifically targeted without due regard for household power dynamics (Fisher and Carr, 2015; Voss *et al.*, 2023a). In practice, this could involve including both spouses in the informed consent process in spousal-couple households to ensure mutual agreement regarding participation. The informed consent process also provides an opportunity to communicate project goals and the rationale for targeting women plot managers, thereby promoting transparency and household support for participation.

However, gender represents only one axis of social difference influencing varietal preferences, production activities, farming experiences, and outcomes (Teeken *et al.*, 2021; Bacud *et al.*, 2024). Important sources of variation include age, marital status, education, land availability, market orientation, social capital, and intra-household dynamics (Acevedo *et al.*, 2020; Nanyonjo *et al.*, 2024; Yami *et al.*, 2024; Nabateregga *et al.*, 2025). An example of the latter is maize production, where joint decision-making within households is common and decisions, including seed choice, are often negotiated rather than made independently (Fisher and Carr, 2015; Voss *et al.*, 2023b). Given this complexity, more nuanced and context-specific approaches should be used to recruit diverse women and men as host farmers.

To operationalize a more inclusive approach to host farmer sampling, typology delineation has been proposed as a sampling tool (Cairns *et al.*, 2026). A typology is a classification of farms into relatively homogeneous groups based on shared socio-economic, biophysical, or management characteristics. Typologies are widely used to differentiate groups of farms and to manage the heterogeneity of smallholder farming systems (Alvarez *et al.*, 2018; Hammond *et al.*, 2020; Manners *et al.*, 2025). Their applications in agronomic research include many purposes, such as understanding variation in the adoption of new technologies (Mujeyi *et al.*, 2022) and tailoring agronomic recommendations to farms differing in resource-endowments (Berre *et al.*, 2019; Mutsamba-Magwaza *et al.*, 2025). Typology delineation could be applied as an intersectional approach to guide the recruitment of host farmers and ensure representativeness of the wider farming population within the TPE. Farm types are delineated within typologies based on farm characteristics. To collect farm characteristics, surveys are conducted with farmers within the TPE. Surveys aim to collect information on variables associated with variation in farming systems and crop production (Fig. 2), including demographic characteristics (particularly farm household head and plot manager genders), farm and livestock endowments, and agronomic management practices (Hassall *et al.*, 2023). The resulting data are used to construct a farm typology and delineate distinct farm types. A stratified sampling approach using region and farm type as strata can be applied to select

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representative host farm (Baudron *et al.*, 2024). Host farmers can then be sampled proportionally from each farm type to reflect their relative prevalence within the target population. Where breeding programs seek to better understand the preferences of specific groups that are underrepresented in the population, particular farm types may be deliberately oversampled. Earlier on-farm trials focused on gender-differentiation at the household level, unintentionally neglecting the experience of women farmers living in households headed by men (Voss *et al.*, 2021). In Southern Africa, farms often contain several plots of maize. Within a household, intrahousehold decision making has been shown to prioritize inputs and labor toward the higher yield fields which tend to be male or jointly managed plots (Farnworth *et al.*, 2017). As a result, women-managed plots often receive less inputs, further reducing fertility, yield, and income. Recognizing the need to include the production environments and preferences of women farmers living within households headed by men, within farm types, targets can should be additionally set for the inclusion of female, male, and joint maize plot managers.

However, there are current limitations in the construction and delineation of farm types to aid host farmer sampling that may limit widescale deployment across CGIAR-NARES breeding networks. Conducting large scale surveys on an annual basis is expensive and requires farmer's valuable time (Tabe-Ojong *et al.*, 2026). Currently, farmers are often asked over 20 questions which are subsequently used to delineate farm types. There is a need to determine the minimum set of variables required to delineate farm types across crops and geographies. Classification random forest analysis has been used to identify the most discriminating variables and guide the selection of key variables for subsequent surveys (Cairns *et al.*, 2026). AI-based methods are well-suited to reducing the dimensionality of farm characterization data while retaining the information required for farm type delineation. Beyond variable selection, AI offers potential to identify those farm characteristics that best explain differences in varietal performance and farmer preferences across production environments. This would allow farm typologies to be constructed around farmer characteristics most relevant to breeding, rather than simply those that best distinguish farm groups. The identification of nonlinear relationships and complex interactions among large numbers of variables is a well-established application of machine learning methods.

Sparse testing strategies to facilitate on-farm testing

Sparse testing is a testing strategy in which only a subset of candidates are evaluated within each environment (Jarquin *et al.*, 2020). Unlike conventional experimental designs, where all candidate entries are tested together at each location with adequate replication to estimate genetic merit, sparse testing allows breeding programs to distribute testing resources across a larger number of environments within the TPE. As a result, candidate varieties can be evaluated directly on farms at scale, which is difficult to achieve with conventional experimental designs. Even at later stages of testing, when only a limited number of highly selected candidates are evaluated, conventional experimental designs require relatively large areas of land, limiting both the number of candidates that can be evaluated on individual farms and the extent of testing across the TPE (Atlin *et al.*, 2001; Setimela *et al.*, 2017; Worku *et al.*, 2020; de Sousa *et al.*, 2024). While the average farm size is 56 ha in Europe and 447 ha in the USA, the average farm size in SSA is often as small as 0.49 ha in Kenya and 0.9 ha in Ethiopia (Rapsomanikis, 2015). Larger trials thus have the potential to impose onerous labor requirements from host farmers (Atlin *et al.*, 2001) and inherently exclude farmers with limited land from participating (de Sousa *et al.*, 2024; Voss *et al.*, 2025).

Sparse testing strategies can be applied at both early- and late-stage testing within breeding pipelines. Implemented as a farm-as-incomplete-block (FAIB) designs (Atlin *et al.*, 2001), sparse testing allows candidates to be evaluated by land-poor farmers and provide a more comprehensive sampling of the TPE. This facilitates the selection of crossing parents for population improvement and candidate varieties for product development based on performance directly within the TPE.

Late-stage on-farm sparse testing

Late-stage on-farm testing moves the last stage of the breeding pipeline into farmers' fields, incorporating candidate variety performance within the TPE and farmer preferences into the final selection process. Several features of sparse testing make it particularly well-suited to scale late-stage testing on-farm. First, the smaller land requirement of sparse testing strategies makes it accessible to smallholder farmers who cannot accommodate a complete block of all candidates, while the overlapping incomplete block design across many farms creates the connectivity needed for robust statistical inference. Second, moving the final stage of testing into the environments in which new varieties must perform increases external validity, captures G×E×M interactions, and may elucidate trait tradeoffs that are important to farmers (Angudubo *et al.*, 2026). Building on this, if routinely implemented within breeding pipelines rather than applied in an ad-hoc manner, late-stage testing in farmers' fields can identify 'spoiler criteria' which could ultimately result in the dis-adoption of varieties (Misiko, 2013). Finally, by reducing the number of entries tested per farm, testing networks using sparse testing can increase sampling of the TPE within a similar budget compared to fully replicated trials.

Given these advantages, sparse late-stage testing has been a key component of efforts to systematically operationalize on-farm testing within breeding pipelines. This shift is reflected in the scale of recent deployment. Late-stage sparse testing in maize breeding pipelines in eastern and southern Africa increased testing from fewer than 20 farms per breeding pipeline

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(Setimela *et al.*, 2012, 2017; Worku *et al.*, 2020) to almost 300 farms (Prasanna *et al.*, 2022; Boddupalli, 2024). One approach that has driven most of this expansion is the triadic comparison of technology options (tricot) approach (van Etten *et al.*, 2019), which has enabled moving late-stage testing from multi-environment trials to farmers' fields in cassava (*Manihot esculenta L.*) (Nanyonjo *et al.*, 2024), common bean (*Phaseolus vulgaris L.*) (Kadege *et al.*, 2026), groundnut (*Arachis hypogaea L.*) (Daudi *et al.*, 2026), maize (*Zea mays L.*) (Voss *et al.*, 2023a), amaranth (*Amaranthus spp.*) (Voss *et al.*, 2025), potato (*Solanum tuberosum L.*) (Kang *et al.*, 2025; Sharma *et al.*, 2025) and yam (*Dioscorea spp.*) (Kalu *et al.*, 2023). Tricot is a citizen science-based approach using an incomplete block design with a block size of three (van Etten *et al.*, 2019). Each farmer ranks three randomly assigned candidates for agronomic performance, end-user traits and overall performance. Individual rankings are statistically aggregated across hundreds of farms to produce a preference scale for all tested candidates (van Etten *et al.*, 2019). While tricot was originally developed for farmer rankings, it can also incorporate quantitative performance data, combining objective measurements with farmer preferences (de Sousa *et al.*, 2024).

Early-stage on-farm sparse testing

Early-stage on-farm sparse testing leverages genomic relationships to evaluate selection candidates within the TPE, ensuring that selection environments reflect the diverse conditions in farmers' fields while incorporating farmers' trait preferences already at early stages of the breeding process.

Early-stage testing within a conventional breeding program is typically characterized by large numbers of genotypes that are evaluated with no or minimal replication. Due to the logistical challenges of phenotyping hundreds to thousands of genotypes, selection pressure is usually high, often exceeding 90% of candidates culled and 10% retained across all trait criteria (Prasanna *et al.*, 2022; Werner *et al.*, 2025). While this is an efficient strategy for eliminating inferior genotypes through rapid phenotypic screening, selection accuracy among the remaining candidates is generally low, creating a strong early-stage selection bottleneck.

Traditionally this selection bottleneck has been managed by concentrating early-stage testing at research stations, where the requirement for comparability can be met. Without genetic relationships linking genotypes across environments, valid comparisons require all candidates to be evaluated under identical or similar conditions. Expanding the number of testing environments is only possible if replication increases proportionally, which quickly becomes logistically and financially prohibitive. Early-stage testing has therefore been historically conducted using a few managed environment trials to represent the TPE (Masuka *et al.*, 2017; Setimela *et al.*, 2017; Prasanna *et al.*, 2022). While this approach guarantees comparability, it comes at the cost of evaluating genotypes in a small number of environments that may not adequately represent the TPE.

Although genomics-based sparse testing is not a novel concept (Burgueño *et al.*, 2012; Lorenz, 2013; Endelman *et al.*, 2014), advances in quantitative genetics, genomics, and trial design facilitate its application in early-stage testing on-farm. The challenge of comparing candidates never evaluated in the same environment is one animal breeding has confronted for decades. Livestock are phenotyped within a single herd and never replicated across farms, so genetic evaluation has always required genetic connectedness across herds, traditionally established through pedigree relationships (Henderson, 1973). Genomic information later strengthened this connectedness, enabling genetic evaluation in smallholder systems where individual farmers only kept a small number of animals (Powell *et al.*, 2021). The same logic is now being applied in crop breeding.

In crops, Endelman *et al.* (2014) were the first to propose sparse testing, showing that a genomic relationship matrix (GRM) can be used to connect unreplicated genotypes across environments. This approach is now widely used in multi-environment trials within breeding pipelines (Jarquin *et al.*, 2020; Atanda *et al.*, 2021; González-Diéguez *et al.*, 2026) and was recently extended to early-stage on-farm testing (Werner *et al.*, 2025). Rather than replicating the selection candidate itself, the GRM leverages marker haplotypes shared among related family members as partial replicates across farms, enabling genomic prediction of performance across the entire TPE. Early-stage on-farm testing complements rather than replaces on-station testing. Yield performance under farmer-managed conditions is best captured in farmers' fields, while traits such as specific disease resistance are more efficiently assessed on-station through controlled inoculation and managed stress conditions, with on-farm trials informing the design of these managed environments in return (Werner *et al.*, 2025). Early-stage on-farm sparse testing is currently being piloted in maize, sorghum, and common bean.

While sparse testing at both early and late stages within breeding pipelines provides greater sampling of the TPE and the inclusion of a diverse range of farmers, these approaches have tradeoffs. In contrast to fully replicated designs, sparse testing does not allow farmers to observe all the genetic diversity and thus could be considered to undermine farmers agency (Oberson *et al.*, 2024). In addition, the limited farmer-researcher interaction provides limited insight into the reasoning behind farmers' preferences compared to farmer research networks and farmer field schools in participatory plant breeding.

Advances and potential bottlenecks in on-farm phenotyping

Phenotyping is critical to accurately estimate the genetic merit of individuals and identify parents for recycling and selection candidates for advancement (Araus *et al.*, 2018). Recent

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advances in digital high-throughput phenotyping and remote sensing are reducing the cost and logistical burden of phenotyping in farmers' fields, making large-scale on-farm testing networks increasingly feasible. While cost effective high throughput phenotyping has been the focus of intensive research to increase genetic gain in crop breeding programs (Reynolds *et al.*, 2020; Araus *et al.*, 2022; Gano *et al.*, 2024; Hoyos-Villegas *et al.*, 2025), it has primarily focused on managed environment trials with limited emphasis on on-farm trials which bring different requirements and challenges. In particular, distance between individual farms is often extensive and, in many regions of the world, travel between farms can be slow due to poor road networks/infrastructure making researcher-led phenotyping resource intensive and a key cost driver (de Sousa *et al.*, 2024).

Yield is generally a must-have trait within most target product profiles, and its selection accuracy is therefore a primary concern. Researcher-led yield measurements are generally considered to have higher accuracy than farmer-led measurements and so have been primarily used on-farm (Setimela *et al.*, 2017; Collinson *et al.*, 2022). However, budget is a major constraint within breeding programs (Chaingeni *et al.*, 2025; Das *et al.*, 2025) and the high cost per data point of researcher-led measurements potentially limits the expansion of on-farm testing networks. For example, in Hamadziripi *et al.* (2025), the cost of researcher-led measurements of grain yield for only 20 locations in one district of Zimbabwe were estimated at 3467 USD, with fuel being the primary cost driver. Cost per data point was the determining factor in limiting trials to a single district.

Given these cost constraints, farmer or agricultural extension agent-led data collection would be more resource-efficient. However, this approach raises concerns about data accuracy (Soleri and Smith, 2002; Kosmowski *et al.*, 2021). Several studies have addressed this concern directly. Nabateregga *et al.* (2025) compared the accuracy of farmer, agricultural extension agent, and researcher-led quantification of common bean yield in Tanzania. Farmers and extension agents were trained prior to yield estimation using commonly available objects (*sado* tins) as a measure. Farmers were encouraged to sun-dry beans for two days after harvest to systematically reduce moisture content. Yield measured by all three groups was highly correlated, indicating high accuracy across groups. Similarly, Ulzen *et al.* (2024) compared four different methods to estimate yield in maize and cocoa (*Theobroma cacao* L.) in farmers' fields in Ghana (ranking sacks by lifting, visual estimates of heap weight, volumetric measurement using a standard tomato tin can, and weighing using a butcher's scale). Of these methods volumetric measurement using a standard tomato tin can was found to have the lowest error measurement, with accuracy largely unaffected by farmer gender, education, eyesight, age, or experience.

Beyond improvements in manual measurement, remote sensing offers a promising avenue for cheap and fast phenotyping in farmers' fields. The extensive proliferation of smart phones worldwide offers options to apply proximal sensing in on-farm testing networks (Araus *et al.*, 2026). For example, image analysis software supported by ImageJ (Makanza *et al.*, 2018) has been used in maize on-farm trials to estimate yield components (ear length, kernel length and width) (Hamadziripi *et al.*, 2025) and fall army worm ear damage (Baudron *et al.*, 2019). Images taken by extension agents are shared via *Whatsapp* and subsequently analyzed by researchers. Similarly, Lasdun *et al.* (2024) developed an image analysis-based tool to estimate common bean bacterial blight severity within on-farm trials. Estimates of disease severity using image analysis of mobile-phone images had a higher level of accuracy relative to actual disease severity when every plot was measured by hand. This overcomes logistical constraints associated with estimating disease prevalence in on-farm trials and, importantly, subjectivity associated with visual estimates (Bock *et al.*, 2022). Building on this, digital phenotyping tools to estimate stand count and pod count in common bean on-farm trials are currently being validated in almost 500 farmers' fields (Lasdun *et al.*, 2024). Beyond trait estimation, images taken in the field can also be used to infer agronomic management practices used by host farmers.

At a larger scale, advances in spatial resolution and revisiting times (the time between satellite passes to observe the same location) are opening new opportunities to leverage satellite-derived indices within on-farm trial networks. While there has been extensive research to estimate traits using unoccupied aerial vehicles (UAVs) equipped with sensors (Hoyos-Villegas *et al.*, 2025), UAVs do not completely overcome the logistical challenges of visiting on-farm trials spread over large geographic areas and complex legal requirements often required to operate UAVs (Shrestha *et al.*, 2025). Satellites, by contrast, offer a more scalable alternative. Satellite-derived indices have been extensively used in precision farming as proxies for nitrogen content and yield estimations (Vallentin *et al.*, 2022). Advances in miniaturization and reduction in launch costs have facilitated the development of new commercial satellites of very high resolution (VHR), while satellite constellations (i.e. a network of similar satellites) enable larger coverage and higher revisiting times for temporal data (Peter *et al.*, 2020). Examples of VHR satellites constellations include SkySat (0.5 m/pixel) and Pelican (Gen-1 at 0.5 m/pixel; Gen-2 at 0.3 m/pixel), both from Planet Labs; WorldView-2 (0.46 m/pixel) and WorlView-3 (0.31 m/pixel) from Maxar; and Pleiades Neo (0.3 m/pixel) from Airbus (Pinto *et al.*, 2023).

Even with these higher resolutions, there remains some challenges, in terms of the current resolution, in the discrimination of pixels within the plot boundaries that are suitable for analysis. While accurate alignments between the images and the vector representing plot boundaries are essential, the probability of having mixed pixels containing information of adjacent plots or soil is still high. A promising approach to this problem was proposed by Victor *et al.* (2024), who trained spatial deep learning models to incorporate contextual

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information surrounding individual plots (i.e. intra-plot information) for predicting phenotypic traits in canola (*Brassica napus* L), including plant height and days to flowering. Beyond these spatial constraints, a second source of difficulty remains in the consistency of temporal phenotypic data derived from satellites, as this depends on calibration protocols that properly account for the atmospheric conditions during the data capture, and the sensor's optical and electronic properties (Saunier *et al.*, 2022). This is particularly relevant in locations where varying meteorological conditions are expected and in situations where high frequency in data collection is only achievable using different sensors from a satellite constellation, where even subtle differences in optics and view angles can result in additional sources of uncertainties.

Combined, these advances provide potential to expand the use of satellite-derived images within on-farm testing using small plots over a large geographic area. However, realizing this potential will depend on the alignment of estimated traits derived from satellite images with the primary needs and requirements of breeding programs. Breeding programs are guided by target product profiles which define the traits a new variety must possess, and the threshold levels of each trait required to meet or exceed the needs of farmers, processors, and consumers within a given crop market segment. Advancement decisions are based on the traits with target product profiles. Thus, it is critical that phenotyping research focused on estimating traits within on-farm testing aligns with trait requirements.

An additional factor in the adoption of VHR satellite imagery for phenotyping is cost. At present, on-demand VHR imagery typically costs several hundred US dollars per image, while historical images may be available at a lower cost but without guarantee of spatial or temporal coverage of specific sites. Moreover, a single acquisition generally covers several square kilometers (e.g. approximately 25 km^2 for a SkySat scene), which is substantially larger than the area required for one or few breeding trials. In addition to this, the processing and management of this imagery requires specific expertise and facilities that come at an extra cost. Taken together, these financial constraints suggest that large-scale adoption of satellite-based phenotyping would depend on coordinated approaches, where organizations such as the CGIAR and NARES align to reduce costs through joint acquisitions, shared infrastructure, and harmonized protocols. Although commercial licensing of VHR imagery often restricts direct data sharing, consortia agreements can facilitate collective access under negotiated terms. Alternatively, collaborative networks based on standardized image-processing workflows could enable partners to share derived products and analytical outputs without transferring the original licensed imagery.

Conclusion

The transition toward breakthrough products, which deliver at least 25% improvements in productivity and value in farmers' fields, represents a step change in ambition of public sector breeding programs. Participatory breeding approaches (PPB and PVS) have a proven track record in developing varieties that offer significant benefit in farmers' fields, with strong adoption capacity (Sperling *et al.*, 1993, 2001; Witcombe *et al.*, 1996; Witcombe, 2000). Despite their success, institutionalizing on-farm testing approaches has been slow (McGuire, 2008; Witcombe and Yadavendra, 2014; Ceccarelli and Grando, 2020). However, twelve years after its development, the tricot approach has been widely used for PVS across crop breeding programs in collaboration with over 150 000 farmers (de Sousa *et al.*, 2024). Furthermore, recognizing the need for robust evidence generated under farmer-managed conditions, the CGIAR Breeding for Tomorrow platform is moving toward institutionalizing late stage on-farm testing across CGIAR-NARES breeding programs.

Advances in inclusive farmer sampling strategies can reduce the recruitment bias that has historically limited the representativeness of on-farm technology testing. Ensuring that host farmers reflect the diversity of the TPE, including women plot managers, resource-poor households, and farmers operating under diverse management regimes, and their agronomic management practices, can help generate data with sufficient external validity to support breakthrough product decisions. Similarly, advances in sparse testing strategies can reduce logistical complexities and cost of running on-farm trials. In general, there have been fewer advances in early stage-on-farm testing approaches, but leveraging the GRM can help move early-stage testing on farm, allowing the selection of crossing parents and early-generation candidates based on performance in the environments where genetic gain ultimately matters. This approach primarily facilitates the evaluation of key traits within on-farm environments rather than directly involving farmers in the decision making of early-stage candidate selection, a key component of success in varieties developed through participatory plant breeding (Oberson *et al.*, 2024). Cost per data point of on-farm phenotyping has constrained the scale of on-farm testing. While challenges remain in terms of spatial resolution, temporal coverage, atmospheric calibration, and the cost of proprietary imagery, coordinated approaches across CGIAR and NARES networks could accelerate access to satellite-based phenotyping in ways that individual programs cannot achieve alone.

On-farm testing networks also provide an opportunity to integrate genetic and agronomic improvement through beneficial G×M interactions. Sub-optimal N fertilization is the largest driver of low maize yields in SSA (Leonardo *et al.*, 2026). Despite the African Union Abuja Declaration to increase fertilizer use to 50 kg N ha^{-1} by 2015, fertilizer use has only increased from 8 kg N ha^{-1} to 18 kg N ha^{-1} over the past two decades (Rasche and Vanlauwe, 2025). Sub-optimal fertilizer use is primarily associated with high farm-gate fertilizer prices (Bonilla-Cedrez *et al.*, 2021) and low and variable economic return on investment in fertilizer

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use (Cairns *et al.*, 2021). The widespread scaling of seed-based technologies in SSA have proved easier than agronomic management practices including digital-based advisories (Sida *et al.*, 2023). Within the context of G×E×M interactions, in rainfed systems, while the E is largely uncontrollable, M, in terms of N application, is theoretically controllable. On-farm testing networks can be exploited to dissect G×N interactions and increase gain through beneficial G×M interactions (Gaffney *et al.*, 2015; Cooper *et al.*, 2021, 2023). Over the past decade, geopolitical shocks have repeatedly disrupted a highly concentrated global fertilizer market, driving price volatility. Farmer recommendations on agronomic management practices to date have generally been static, overlooking market conditions which affect the financial viability of different agronomic recommendations. By concurrently quantifying farm-gate fertilizer prices, fertilizer use, and agronomic management practices of host farmers, economic returns on fertilizer use efficiency could be determined in a volatile and changing market to provide farmers more up-to-date recommendations (Bonilla-Cedrez *et al.*, 2021).

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Author contributions

JEC and CRW: conceptualization, JEC, CRW, MF, FP: writing—original draft, KdS, JvE, HD, MN, MZA: review and editing.

Conflict of interest

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References

Acevedo M, Pixley K, Zinyengere N, Meng S, Tufan H, Cichy K, Bizikova L, Isaacs K, Ghezzi-Kopel K, Porciello J. 2020. A scoping review of adoption of climate-resilient crops by small-scale producers in low- and middle-income countries. Nature Plants **6**, 1231–1241.

Ae-Ngibise KA, Asare-Doku W, Peprah J, Mujtaba MN, Nifasha D, Donnir GM. 2021. The mental health outcomes of food insecurity and insufficiency in West Africa: a systematic narrative review. Behavioural Sciences **11**, 146.

Alvarez S, Timler CJ, Michalscheck M, Paas W, Descheemaeker K, Tittonell P, Andersson JA, Groot JCJ. 2018. Capturing farm diversity with hypothesis-based typologies: an innovative methodological framework for farming system typology development. PLoS One **13**, e0194757.

Amondo E, Simtow F, Rahut DB, Erenstein O. 2019. Productivity and production risk effects of adopting drought-tolerant maize varieties in Zambia. International Journal of Climate Change Strategies and Management **11**, 570–591.

Angudubo S, Benavente G, van Etten J, Sinyinda L. 2026. Insights into the costing of on-farm testing: a case study of maize tricot trials by Zambia agricultural and research institute (ZARI). Kampala, Uganda: Biodiversity International and the International Center for Tropical Agriculture. https://alliancebioversityciat.org/publications-data/insights-costing-farm-testing-case-study-maize-tricot-trials-zambia-agricultural

Aramburu-Merlos F, Tenorio FAM, Mashingaidze N, Sananka A, Aston S, Ojeda JJ, Grassini P. 2024. Adopting yield-improving practices to meet maize demand in sub-saharan Africa without cropland expansion. Nature Communications **15**, 4492.

Araus JL, Jauregui-Besó J, Segarra J, Junliang F. 2026. Affordable crop phenotyping matters. Crop and Environment **5**, 100145.

Araus JL, Kefauver SC, Vergara-Díaz O, *et al.* 2022. Crop phenotyping in a context of global change: what to measure and how to do it. Journal of Integrative Plant Biology **64**, 592–618.

Araus JL, Kefauver SC, Zaman-Allah M, Olsen MS, Cairns JE. 2018. Translating high-throughput phenotyping into genetic gain. Trends in Plant Science **23**, 451–466.

Atanda SA, Olsen M, Crossa J, *et al.* 2021. Scalable sparse testing genomic selection strategy for early yield testing stage. Frontiers in Plant Science **12**, 658978.

Atlin GN, Cooper M, Bjørnstad Å. 2001. A comparison of formal and participatory breeding approaches using selection theory. Euphytica: Netherlands Journal of Plant Breeding **122**, 463–475.

Bacud EST, Gerullis MK, Puskur R, Heckelei T. 2024. Looking at gender is not enough—how diversity of farmers' marginalization relates to varietal trait preferences. Food Policy **124**, 102616.

Baldos ULC, Cisneros-Pineda A, Fuglie KO, Hertel TW. 2025. Adoption of improved crop varieties limited biodiversity losses, terrestrial carbon emissions, and cropland expansion in the tropics. Proceedings of the National Academy of Sciences, USA **122**, e2404839122.

Balogun I, Garner E, Amer P, *et al.* 2022. From traits to typologies: piloting new approaches to profiling trait preferences along the cassava value chain in Nigeria. Crop Science **62**, 259–274.

Bänziger M, de Meyer J. 2002. Maize variety development for stress-prone environments in Southern Africa. In: Cleveland D, Soleri D, eds. Farmers, scientists and plant breeding: integrating knowledge and practice. Wallingford: CABI, 269–296.

Bänziger M, Diallo A. 2001. Progress in developing drought and N stress tolerant maize cultivars for eastern and Southern Africa. In: Friesen DK, Palmer AFE, eds. Proceedings of the 7th eastern and Southern Africa regional maize conference. Nairobi, Kenya: CIMMYT and KARI, 189–194.

Bänziger M, Setimela PS, Hodson D, Vivek B. 2006. Breeding for improved abiotic stress tolerance in maize adapted to Southern Africa. Agricultural Water Management **80**, 212–224.

Baudron F, Cairns JE, Haefele SM, Hassall KL, Ndhlela T, Nyagumbo I, Manzeke-Kangara G, Joy EJM. 2024. Projecting the contribution of pro-vitamin A maize biofortification and other nutrition interventions to the nutritional adequacy and cost of diets in rural Zimbabwe. Journal of Nutrition **154**, 1815–1826.

Baudron F, Zaman-Allah MA, Chaipa I, Chari N, Chinwada P. 2019. Understanding the factors influencing fall armyworm (spodoptera frugiperda J.E. Smith) damage in African smallholder maize fields and quantifying its impact on yield. A case study in eastern Zimbabwe. Crop Protection **120**, 141–150.

Berre D, Baudron F, Kassie M, Craufurd P, Lopez-Ridaura S. 2019. Different ways to cut a cake: COMPARING expert-BASED and statistical typologies to target sustainable intensification technologies, A case-study in southern Ethiopia. Experimental Agriculture **55**, 191–207.

Bock CH, Chiang K-S, Del Ponte EM. 2022. Plant disease severity estimated visually: a century of research, best practices, and opportunities for

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improving methods and practices to maximize accuracy. Tropical Plant Pathology **47**, 25–42.

Boddupalli PM. 2024. ABI End of initiative–2024 progress deep dive of the maize breeding teams from CIMMYT and IITA. Nairobi, Kenya: CIMMYT.

Bolaños J, Edmeades GO. 1993. Eight cycles of selection for drought tolerance in lowland tropical maize. I. Responses in grain yield, biomass, and radiation utilization. Field Crops Research **31**, 233–252.

Bonilla-Cedrez C, Chamberlin J, Hijmans RJ. 2021. Fertilizer and grain prices constrain food production in sub-saharan Africa. Nature Food **2**, 766–772.

Burgueño J, de los Campos G, Weigel K, Crossa J. 2012. Genomic prediction of breeding values when modeling genotype × environment interaction using pedigree and dense molecular markers. Crop Science **52**, 707–719.

Burrows K, Denckla CA, Hahn J, Schiff JE, Okuzono SS, Randriamady H, Mita C, Kubzansky LD, Koenen KC, Lowe SR. 2024. A systematic review of the effects of chronic, slow-onset climate change on mental health. Nature: Mental Health **2**, 228–243.

Byrne PF, Bolaños J, Edmeades GO, Eaton DL. 1995. Gains from selection under drought versus multilocation testing in related tropical maize populations. Crop Science **35**, 63–69.

Cai W, Reason C, Mohino E, *et al.* 2025. Climate impacts of the el niño–southern oscillation in Africa. Nature Reviews: Earth & Environment **6**, 503–520.

Cairns JE, Chamberlin J, Rutsaert P, Voss RC, Ndhlela T, Magorokosho C. 2021. Challenges for sustainable maize production of smallholder farmers in sub-Saharan africa. Journal of Cereal Science **101**, 103274.

Cairns JE, Crossa J, Zaidi PH, *et al.* 2013. Identification of drought, heat, and combined drought and heat tolerant donors in maize. Crop Science **53**, 1335–1346.

Cairns JE, Prasanna BM. 2018. Developing and deploying climate-resilient maize varieties in the developing world. Current Opinion in Plant Biology **45**, 226–230.

Cairns JE, Zaman-Allah M, Baudron F, Voss RC, Hassall KL, Euler M, Amondo E, Marenya PP, Prasanna BM. 2026. Know your men and women farmers: ensuring host farmers in participatory trials represent heterogeneity within the target environment. Plants, People, Planet **8**, 259–276.

Ceccarelli S. 1989. Wide adaptation: how wide? Euphytica: Netherlands Journal of Plant Breeding **40**, 197–205.

Ceccarelli S. 1994. Specific adaptation and breeding for marginal conditions. Euphytica: Netherlands Journal of Plant Breeding **77**, 205–219.

Ceccarelli S, Grando S. 2020. Participatory plant breeding: who did it, who does it and where? Experimental Agriculture **56**, 1–11.

Ceccarelli S, Grando S, Bailey E, Amri A, El-Felah M, Nassif F, Rezgui S, Yahyaoui A. 2001. Farmer participation in barley breeding in Syria, Morocco and Tunisia. Euphytica: Netherlands Journal of Plant Breeding **122**, 521–536.

Chaingeni D, Mukaro R, Sneller C, *et al.* 2025. More bang for your buck: potential gains through optimizing maize breeding schemes in sub-saharan Africa. Frontiers in Plant Science **16**, 1553272.

Chenu K, Cooper M, Hammer GL, Mathews KL, Dreccer MF, Chapman SC. 2011. Environment characterization as an aid to wheat improvement: interpreting genotype-environment interactions by modelling water-deficit patterns in north-eastern Australia. Journal of Experimental Botany **62**, 1743–1755.

Chikowo R, Zingore S, Snapp S, Johnston A. 2014. Farm typologies, soil fertility variability and nutrient management in smallholder farming in subsaharan Africa. Nutrient Cycling in Agroecosystems **100**, 1–18.

Chivasa W, Worku M, Teklewold A, Setimela P, Gethi J, Magorokosho C, Davis NJ, Prasanna BM. 2022. Maize varietal replacement in eastern and Southern Africa: bottlenecks, drivers and strategies for improvement. Global Food Security **32**, 100589.

CIMMYT. 1989. An account of how priorities are set among mega-environments from a breeding perspective. Mexico D.F.: CIMMYT.

Coaldrake P, Werner C. 2025. Why CGIAR crop breeding is shifting focus to breakthrough products. https://www.cgiar.org/news-events/news/why-cgiar-crop-breeding-is-shifting-focus-to-breakthrough-products

Collinson S, Hamadzipiri E, De Groote H, Ndegwa M, Cairns JE, Albertsen M, Ligeyo D, Mashingaidze K, Olsen M. 2022. Incorporating male sterility increases hybrid maize yield in low input African farming systems. Communications Biology **5**, 729.

Comstock RE. 1977. Quantitative genetics and the design of breeding programs, eds. Proceedings of the international conference on quantitative genetics. Ames: Iowa State University Press, 1705–1718.

Conti C, Zanello G, Hall A. 2021. Why are agri-food systems resistant to new directions of change? A systematic review. Global Food Security **31**, 100576.

Cooper M, Gho C, Leafgren R, Tang T, Messina C. 2014. Breeding drought-tolerant maize hybrids for the US corn-belt: discovery to product. Journal of Experimental Botany **65**, 6191–6204.

Cooper M, Messina CD. 2023. Breeding crops for drought-affected environments and improved climate resilience. The Plant Cell **35**, 162–186.

Cooper M, Messina CD, Tang T, *et al.* 2022. Predicting genotype × environment × management (G × E × M) interactions for the design of crop improvement strategies: integrating breeder, agronomist, and farmer perspectives. In: Goldman I, ed. Plant breeding reviews, 1st ed. Wiley, Hoboken, USA, 467–585.

Cooper M, Powell O, Gho C, Tang T, Messina C. 2023. Extending the breeder's equation to take aim at the target population of environments. Frontiers in Plant Science **14**, 1129591.

Cooper M, Voss-Fels KP, Messina CD, Tang T, Hammer GL. 2021. Tackling G × E × M interactions to close on-farm yield-gaps: creating novel pathways for crop improvement by predicting contributions of genetics and management to crop productivity. TAG: Theoretical and Applied Genetics: Theoretische Und Angewandte Genetik **134**, 1625–1644.

Das B, Mutiga SK, Odiyo O, *et al.* 2025. Costing of the breeding operations for the national maize programs in eastern and Southern Africa. Frontiers in Sustainable Food Systems **9**, 1545600.

Daudi H, Bujiku AM, Oteng-Frimpong R, *et al.* 2026. Analysis of genotype-by-environment interaction, farmers' preferences, and gender response in groundnut breeding in Tanzania using triadic comparison of technologies. Food and Energy Security **15**, e70228.

de Roo N, Andersson JA, Krupnick TJ. 2019. On-farm trials for development impact? The organisation of research and the scaling of agricultural technologies. Experimental Agriculture **55**, 163–184.

de Sousa K, van Etten J, Manners R, *et al.* 2024. The tricot approach: an agile framework for decentralized on-farm testing supported by citizen science. A retrospective. Agronomy for Sustainable Development **44**, 8.

Donovan J, Rutsaert P, Mawia H, de Sousa K, van Etten J. 2025. Farmers' preferences for the next generation of maize hybrids: application of product concept testing in Kenya and Uganda. Experimental Agriculture **61**, e5.

Doss C. 2013. Intrahousehold bargaining and resource allocation in developing countries. The World Bank Research Observer **28**, 52–78.

Dowd-Uribe B, Blundo-Canto G, Glover D, Louafi S, Shilomboleni H, Rock JS, Kikulwe EM, Fischer K, Joly P-B. 2024. Socio-economic assessment and genetically engineered crops in Africa: building knowledge for development? Global Food Security **42**, 100782.

Edmeades GO, Bolafios J, Lafitte HR, Rajaram S, Pfeiffer W, Fischer RA. 1989. Traditional approaches to breeding for drought resistance in cereals. In: Baker FWG, ed. Drought resistance in cereals. Wallingford: ICSU and CABI, 27–52.

Endelman JB, Atlin GN, Beyene Y, Semagn K, Zhang X, Sorrells ME, Jannink JL. 2014. Optimal design of preliminary yield trials with genome-wide markers. Crop Science **54**, 48–59.

Euler M, Cairns JE, Ngoma H, Manyanga M, Zaman-Allah M. 2026. Stress-tolerant maize under drought: heterogeneous effects on productivity and food security in Zimbabwe. Food Security. doi: 10.1007/s12571-026-01709-x

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Farnworth CR, Stirling C, Sapkota TB, Jat ML, Misiko M, Attwood S. 2017. Gender and inorganic nitrogen: what are the implications of moving towards a more balanced use of nitrogen fertilizer in the tropics? International Journal of Agricultural Sustainability **15**, 136–152.

Fischer KS, Palmer AFE. 1984. Tropical maize. In: Goldsworthy PR, Fischer NM, eds. The physiology of tropical field crops. New York: Wiley, 213–248.

Fisher M, Carr ER. 2015. The influence of gendered roles and responsibilities on the adoption of technologies that mitigate drought risk: the case of drought-tolerant maize seed in eastern Uganda. Global Environmental Change: Human and Policy Dimensions **35**, 82–92.

Freeman HA. 2001. Comparison of farmer-participatory research methodologies: case studies in Malawi and Zimbabwe. Nairobi, Kenya: Socioeconomics and Policy Program, International Crops Research Institute for the Semi-Arid Tropics.

Gaffney J, Schussler J, Löffler C, Cai W, Paszkiewicz S, Messina C, Groeteke J, Keaschall J, Cooper M. 2015. Industry-scale evaluation of maize hybrids selected for increased yield in drought-stress conditions of the US corn belt. Crop Science **55**, 1608–1618.

Gano B, Bhadra S, Vilbig JM, Ahmed N, Sagan V, Shakoor N. 2024. Drone-based imaging sensors, techniques, and applications in plant phenotyping for crop breeding: a comprehensive review. Plant Phenome Journal **7**, e5.

Gebre GG, Lakew HW, Fikadu AA, Jiren TS, Wenda BDS, Löhr K, Sieber S. 2025. Effect of adopting stress-tolerant maize varieties on yields and food security in Kenya. International Journal of Sustainable Development & World Ecology **32**, 1022–1035.

Gebre GG, Mawia H, Makumbi D, Rahut DB. 2021. The impact of adopting stress-tolerant maize on maize yield, maize income, and food security in Tanzania. Food and Energy Security **10**, e313.

González-Diéguez DO, Atlin GN, Beyene Y, Wegary D, Gemenet DC, Werner CR. 2026. Sparse testcrossing for early-stage genomic prediction of general combining ability to increase genetic gain in maize hybrid breeding programs. TAG: Theoretical and Applied Genetics: Theoretische Und Angewandte Genetik **139**, 78.

Hailemichael S, Haug R. 2020. The use and abuse of the 'model farmer' approach in agricultural extension in Ethiopia. The Journal of Agricultural Education and Extension **26**, 465–484.

Hamadziripi ET, Collinson S, Voss RC, Baudron F, Labuschagne MT, Franke AC, Zaman-Allah M, Olsen MS, Burgueño J, Cairns JE. 2025. Validating a novel genetic technology for hybrid maize seed production under management practices associated with resource-poor farmers in zimbabwe. Plants, People, Planet **7**, 801–815.

Hammond J, Rosenblum N, Breseman D, Gorman L, Manners R, van Wijk MT, Sibomana M, Remans R, Vanlauwe B, Schut M. 2020. Towards actionable farm typologies: scaling adoption of agricultural inputs in Rwanda. Agricultural Systems **183**, 102857.

Hartkamp AD, White JW, Rodgriguez Aguilar A, Bänziger M, Srinivasan G, Granados G, Crossa J. 2000. Maize production environments revisited: a GIS-based approach. Mexico D.F., Mexico: International Maize and Wheat Improvement Centre (CIMMYT).

Hassall KL, Baudron F, MacLaren C, Cairns JE, Ndhlela T, McGrath S, Nyagumbo I, Haefele S. 2023. Construction of a generalised farm typology to aid selection, targeting and scaling of onfarm research. Computers and Electronics in Agriculture **212**, 108074.

Heino M, Kinnunen P, Anderson W, Ray DK, Puma MJ, Varis O, Siebert S, Kummu M. 2023. Increased probability of hot and dry weather extremes during the growing season threatens global crop yields. Scientific Reports **13**, 3583.

Henderson CR. 1973. Sire evaluation and genetic trends. Proceedings of the animal breeding and genetics symposium in honor of Dr Jay L. Lush. Champaign, IL: ASAS and ADSA, 10–41.

Hoyos-Villegas V, Jackson M, Vargas-Cedeño M, *et al.* 2025. Affordable phenomics: expanding access to enhancing genetic gain in plant breeding. Plant Phenome Journal **8**, e70034.

Hunt L, Thompson JJ, Niles MT. 2025. How on-farm research project participants compare to a general sample of farmers: a case study of US cover crop farmers. Agronomy for Sustainable Development **45**, 5.

Isaacs K, Weltzien E, Some H, Diallo A, Diallo B, Sidibé M, vom Brocke K, Samake B, Nebié B, Rattunde FW. 2024. Increasing sorghum yields for smallholder farmers in Mali: the evolution towards a context-driven, on-farm, gender-responsive sorghum breeding program. Frontiers in Sustainable Food Systems **8**, 1334385.

Jarquin D, Howard R, Crossa J, *et al.* 2020. Genomic prediction enhanced sparse testing for multi-environment trials. G3 (Bethesda) **10**, 2725–2739.

Kadege EL, Venkataramana PB, Assefa T, Ndunguru JC, Kalemera SM, Ochieng J, Nchanji EB, Rubyogo JC, Mbega ER. 2026. Tanzanian farmers' preferred traits for common bean varieties: breeding insights from the tricot approach. Frontiers in Sustainable Food Systems **9**, 1639680.

Kalu C, Nnabue I, Edemodu A, Agre PA, Adebola P, Asfaw A, Obidiegwu JE. 2023. Farmers' perspective toward a demand led yam breeding in Nigeria. Frontiers in Sustainable Food Systems **7**, 1227920.

Kamali B, Abbaspour KC, Wehrli B, Yang H. 2018. Drought vulnerability assessment of maize in sub-saharan Africa: insights from physical and social perspectives. Global and Planetary Change **162**, 266–274.

Kang G, de Sousa K, Manners R, *et al.* 2025. Integrating environmental, socio-economic, and biological data in a farmer-led potato trial for enhanced varietal assessment in Rwanda. Experimental Agriculture **61**, e18.

Katengeza SP, Holden ST. 2021. Productivity impact of drought tolerant maize varieties under rainfall stress in Malawi: a continuous treatment approach. Agricultural Economics **52**, 157–171.

Kool H, Andersson JA, Giller KE. 2020. Reproducibility and external validity of on-farm experimental research in Africa. Experimental Agriculture **56**, 587–607.

Kosmowski F, Chamberlin J, Ayalew H, Sida T, Abay K, Craufurd P. 2021. How accurate are yield estimates from crop cuts? Evidence from smallholder maize farms in Ethiopia. Food Policy **102**, 102122.

Laajaj R, Macours K, Masso C, Thuita M, Vanlauwe B. 2020. Reconciling yield gains in agronomic trials with returns under African smallholder conditions. Scientific Reports **10**, 14286.

Lasdun L, Güereña D, Ortiz-Crespo B, Mutuvi S, Selvaraj M, Assefa T. 2024. Participatory AI for inclusive crop improvement. Agricultural Systems **220**, 104054.

Leonardo WJ, Silva JV, Nagarajan L, Singh U. 2026. Unravelling the contribution of G × E × M interactions to maize productivity in central Mozambique amid climate change. Experimental Agriculture **62**, e10.

Lesk C, Anderson W, Rigden A, Coast O, Jägermeyr J, McDermid S, Davis KF, Konar M. 2022. Compound heat and moisture extreme impacts on global crop yields under climate change. Nature Reviews: Earth & Environment **3**, 872–889.

Lobell DB, Bänziger M, Magorokosho C, Vivek B. 2011. Nonlinear heat effects on African maize as evidenced by historical yield trials. Nature Climate Change **1**, 42–45.

Lorenz AJ. 2013. Resource allocation for maximizing prediction accuracy and genetic gain of genomic selection in plant breeding: a simulation experiment. G3 (Bethesda) **3**, 481–491.

Lunduka RW, Mateva KI, Magorokosho C, Manjeru P. 2019. Impact of adoption of drought-tolerant maize varieties on total maize production in south eastern Zimbabwe. Climate and Development **11**, 35–46.

Makanza R, Zaman-Allah M, Cairns JE, *et al.* 2018. High-throughput method for ear phenotyping and kernel weight estimation in maize using ear digital imaging. Plant Methods **14**, 49.

Manners R, Hammond J, Umugabe DR, Sibomana M, Schut M. 2025. A farm typology development cycle: from empirical development through validation, to large-scale organisational deployment. Agricultural Systems **224**, 104250.

Downloaded from academic.oup.com/jxb/advance-article/doi/10.1093/jxb/erag431/8785350 by guest on 30 October 2026

Marcos-Garcia P, Carmona-Moreno C, Pastori M. 2024. Intra-growing season dry–wet spell pattern is a pivotal driver of maize yield variability in sub-Saharan Africa. Nature Food **5**, 775–786.

Masuka B, Atlin GN, Olsen M, *et al.* 2017. Gains in maize genetic improvement in eastern and Southern Africa: I. CIMMYT hybrid breeding pipeline. Crop Science **57**, 168–179.

Masvaya EN, Nyamangara J, Nyawasha RW, Zingore S, Delve RJ, Giller KE. 2010. Effect of farmer management strategies on spatial variability of soil fertility and crop nutrient uptake in contrasting agro-ecological zones in Zimbabwe. Nutrient Cycling in Agroecosystems **88**, 111–120.

McGuire SJ. 2008. Path-dependency in plant breeding: challenges facing participatory reforms in the Ethiopian Sorghum improvement program. Agricultural Systems **96**, 139–149.

Misiko M. 2013. Dilemma in participatory selection of varieties. Agricultural Systems **119**, 35–42.

Moyo S. 2000. Zimbabwe environmental dilemma: balancing resource inequities. Zimbabwe Environmental Research Organization. Harare, Zimbabwe.

Mujeyi A, Mudhara M, Mutenje MJ. 2022. Adoption patterns of climate-smart agriculture in integrated crop-livestock smallholder farming systems of Zimbabwe. Climate and Development **14**, 399–408.

Mulungu K, Manning DT, Bozzola M. 2025. Once bitten, twice shy? Direct and indirect effects of weather shocks on fertilizer and improved seeds use. Food Policy **133**, 102852.

Mutsamba-Magwaza EF, Baudron F, Franke AC, Van Der Watt E, Nyagumbo I. 2025. Learning from positive deviants' practices to improve the performance of mixed crop-livestock systems in Zimbabwe. Agricultural Systems **228**, 104397.

Nabateregga M, Dorado-Betancourt H, Solberg SØ, van Etten J, van Heerwaarden J, Gregory T, de Sousa K. 2025. Accuracy of farmer-generated yield estimations of common bean in decentralised on-farm trials in sub–Saharan Africa. European Journal of Agronomy: The Journal of the European Society for Agronomy **170**, 127730.

Nanyonjo AR, Angudubo S, Iragaba P, *et al.* 2024. On-farm evaluation of cassava clones using the triadic comparison of technology options approach. Crop Science **64**, 2679–2697.

Ngoma H, Setimela P, Silva JV, Krishna V. 2025. Poverty and yield effects of CGIAR maize varieties in smallholder farming systems of Zambia. Journal of the Agricultural and Applied Economics Association **4**, 151–166.

Oberson N, Moussa HO, Aminou AM, Kidane YG, Luo JN, Giuliani A, Weltzien E, Haussmann BIG. 2024. Participatory research at scale: a comparative analysis of four approaches to large-scale agricultural technology testing with farmers. Outlook on Agriculture **53**, 320–335.

Ochieng H, Ojiem J, Otieno J. 2019. Farmer versus researcher data collection methodologies: Understanding variations and associated trade-offs. AfricArxiv doi: 10.31730/osf.io/ncw8a. [Preprint].

Ochieng W, Silvert CJ, Diaz J. 2022. Exploring the impacts of lead farmer selection on community social learning: the case of farmer-to-farmer model: a review of literature. Journal of International Agricultural and Extension Education **29**, 7–31.

Peter BG, Messina JP, Carroll JW, Zhi J, Chimonyo V, Lin S, Snapp SS. 2020. Multi-spatial resolution satellite and sUAS imagery for precision agriculture on smallholder farms in Malawi. Photogrammetric Engineering and Remote Sensing **86**, 107–119.

Pinto F, Zaman-Allah M, Reynolds M, Schulthess U. 2023. Satellite imagery for high-throughput phenotyping in breeding plots. Frontiers in Plant Science **14**, 1114670.

Pires CB, Krupek FS, Carmona GI, *et al.* 2024. Perspective of US farmers on collaborative on-farm agronomic research. Agronomy Journal **116**, 1590–1602.

Powell O, Mrode R, Gaynor RC, Johnsson M, Gorjanc G, Hickey JM. 2021. Genomic evaluations using data recorded on smallholder dairy farms in low- to middle-income countries. JDS Communications **2**, 366–370.

Prasanna BM, Burgueño J, Beyene Y, *et al.* 2022. Genetic trends in CIMMYT's tropical maize breeding pipelines. Scientific Reports **12**, 20110.

Rapsomanikis G. 2015. The economic lives of smallholder farmers: an analysis based on household data from nine countries. Rome: Food and Agricultural Organisation of the United Nations.

Rasche F, Vanlauwe B. 2025. From strategy to action: leveraging transformative levers to support Africa's fertilizer and soil health strategy. Soil Security **20**, 100196.

Ray DK, Gerber JS, MacDonald GK, West PC. 2015. Climate variation explains a third of global crop yield variability. Nature Communications **6**, 5989.

Reynolds M, Chapman S, Crespo-Herrera L, *et al.* 2020. Breeder friendly phenotyping. Plant Science **295**, 110396.

Rice B, Kramer B, Trachtman C. 2024. Barriers to using market intelligence in plant breeding: evidence from a survey of breeding professionals. In: Market Intelligence Brief Series 21. Montpellier: CGIAR.

Saunier S, Pflug B, Lobos IM, *et al.* 2022. Sen2Like: paving the way towards harmonization and fusion of optical data. Remote Sensing **14**, 3855.

Schnurr MA, Shilomboleni H, Taylor A, Dowd-Uribe B. 2025. Towards a more pluralistic approach to evaluating the farm-level impacts of new breeding technologies in sub-saharan Africa. Genome **68**, 1–8.

Setimela P, Chitalu Z, Jonazi J, Mambo A, Hodson D, Bänziger M. 2005. Environmental classification of maize-testing sites in the SADC region and its implication for collaborative maize breeding strategies in the subcontinent. Euphytica: Netherlands Journal of Plant Breeding **145**, 123–132.

Setimela P, Zaman-Allah MA, Gasura E, Cairns JE, Thierfelder C, Prasanna BM. 2018. When the going gets tough: performance of stress tolerant maize during the 2015/16 (el niño) and 2016/17 (la niña) season in Southern Africa. Agriculture, Ecosystems & Environment **268**, 79–89.

Setimela PS, MacRobert J, Atlin GN, Magorokosho C, Tarekegne A, Makumbi D. 2012. Evaluation of regional on-farm variety trials in eastern and Southern Africa–2011. Harare: CIMMYT.

Setimela PS, Magorokosho C, Lunduka R, Gasura E, Makumbi D, Tarekegne A, Cairns JE, Ndhela T, Erenstein O, Mwangi W. 2017. On-farm yield gains with stress-tolerant maize in eastern and Southern Africa. Agronomy Journal **109**, 406–417.

Sharma K, Atieno E, Mugo J, de Sousa K, van Etten J, Nyawade S. 2025. Understanding farmer preferences through citizen science: insights on potato varieties in Nigeria. Journal of Agriculture and Food Research **23**, 102135.

Shrestha N, Powadi A, Davis J, *et al.* 2025. Plot-level satellite imagery can substitute for UAVs in assessing maize phenotypes across multistate field trials. Plants, People, Planet **7**, 1011–1026.

Sida TS, Gameda S, Chamberlin J, Andersson JA, Getnet M, Woltering L, Craufurd P. 2023. Failure to scale in digital agronomy: an analysis of site-specific nutrient management decision-support tools in developing countries. Computers and Electronics in Agriculture **212**, 108060.

Soleri D, Smith SE. 2002. Rapid estimation of broad sense heritability of farmer-managed maize populations in the central valleys of oaxaca, Mexico, and implications for improvement. Euphytica: Netherlands Journal of Plant Breeding **128**, 105–119.

Sperling L, Ashby JA, Smith ME, Weltzien E, McGuire S. 2001. A framework for analyzing participatory plant breeding approaches and results. Euphytica: Netherlands Journal of Plant Breeding **122**, 439–450.

Sperling L, Loevinsohn M, Ntabomvura B. 1993. Rethinking the farmer's role in plant breeding: local bean experts and on-station selection in Rwanda. Experimental Agriculture **29**, 509–519.

Tabe-Ojong MP Jr, Abay KA, Abate GT, Machio PM. 2026. When the going gets tough: experimental evidence of respondent fatigue in household surveys. Journal of African Economies **35**, ejag001.

Tall M, Sylla MB, Dajuma A, *et al.* 2023. Drought variability, changes and hot spots across the African continent during the historical period (1928–2017). International Journal of Climatology: A Journal of the Royal Meteorological Society **43**, 7795–7818.

Tarekegne A, Wegary D, Cairns JE, *et al.* 2023. Genetic gains in early maturing maize hybrids developed by the international maize and wheat

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improvement center in Southern Africa during 2000-2018. Frontiers in Plant Science **14**, 1321308.

Technow F, Podlich D, Cooper M. 2026. Back to the future 2: the implications of germplasm structure on the balance between short- and long-term genetic gain in a changing target population of environments. G3 (Bethesda) **16**, jkag044.

Teeken B, Garner E, Agbona A, *et al.* 2021. Beyond "women's traits": exploring how gender, social difference, and household characteristics influence trait preferences. Frontiers in Sustainable Food Systems **5**, 740926.

Tesfaye K, Sonder K, Cairns J, Magorokosho C, Tarekegne A, Kassie GT, Getaneh F, Abdoulaye T, Abate T, Erenstein O. 2016. Targeting drought tolerant maize varieties in Southern Africa: a geospatial crop modelling approach using big data. International Food and Agribusiness Management Review **19**, 75–92.

Thoithi W, Blamey RC, Reason CJC. 2021. Dry spells, wet days, and their trends across Southern Africa during the summer rainy season. Geophysical Research Letters **48**, e2020GL091041.

Thuijsman ES, den Braber HJ, Andersson JA, Descheemaeker K, Baudron F, López-Ridaura S, Vanlauwe B, Giller KE. 2022. Indifferent to difference? Understanding the unequal impacts of farming technologies among smallholders. A review. Agronomy for Sustainable Development **42**, 41.

Toffolini Q, Jeufroy M-H. 2022. On-farm experimentation practices and associated farmer-researcher relationships: a systematic literature review. Agronomy for Sustainable Development **42**, 114.

Ulzen J, Osei-Bonsu NO, Bush G, Betancourt HAD, van Etten J. 2024. Accuracy in agricultural citizen science: farmers' assessment of harvest output in Ghana. Wallingford: CABI. agriRxiv doi: 10.31220/agriRxiv.2025.00295

Vallentin C, Harfenmeister K, Itzerott S, Kleinschmit B, Conrad C, Spengler D. 2022. Suitability of satellite remote sensing data for yield estimation in northeast Germany. Precision Agriculture **23**, 52–82.

van Eeuwijk FA, Cooper M, DeLacy IH, Ceccarelli S, Grando S. 2001. Some vocabulary and grammar for the analysis of multi-environment trials, as applied to the analysis of FPB and PPB trials. Euphytica: Netherlands Journal of Plant Breeding **122**, 477–490.

van Etten J, de Sousa K, Aguilar A, *et al.* 2019. Crop variety management for climate adaptation supported by citizen science. Proceedings of the National Academy of Sciences, USA **116**, 4194–4199.

Victor B, Nibali A, Newman SJ, Coram T, Pinto F, Reynolds M, Furbank RT, He Z. 2024. High-throughput plot-level quantitative phenotyping using convolutional neural networks on very high-resolution satellite images. Remote Sensing **16**, 282.

vom Brocke K, Kondombo CP, Guillet M, Kaboré R, Sidibé A, Temple L, Trouche G. 2020. Impact of participatory sorghum breeding in Burkina Faso. Agricultural Systems **180**, 102775.

Voss RC, Cairns JE, Olsen MS, Muteti FN, Magambo G, Hamadziripi E, Ligeyo D, Mashingaidze K, Collinson S, Woyengo V. 2023a. Innovative approaches to integrating gender into conventional maize breeding: lessons from the seed production technology for Africa project. Frontiers in Sociology **8**, 1254595.

Voss RC, de Sousa K, N'Danikou S, *et al.* 2025. Citizen science informs demand-driven breeding of opportunity crops. Plants, People, Planet **7**, 1700–1713.

Voss RC, Donovan J, Rutsaert P, Cairns JE. 2021. Gender inclusivity through maize breeding in Africa: a review of the issues and options for future engagement. Outlook on Agriculture **50**, 392–405.

Voss RC, Gitonga ZM, Donovan J, Garcia-Medina M, Muindi P. 2023b. Can I speak to the manager? The gender dynamics of decision-making in Kenyan maize plots. Agriculture and Human Values **41**, 205–224.

Werner C, Zaman-Allah M, Mamo TA, Cairns JE, Atlin GN. 2025. Accelerating genetic gain through early-stage on-farm sparse testing. Trends in Plant Science **30**, 17–20.

Witcombe JR. 2000. The impact of decentralized and participatory plant breeding on the genetic base of crops. In: Cooper HD, Spillane C, Hodgkin T, eds. Broadening the genetic base of crop production. Wallingford: CABI, 407–418.

Witcombe JR, Joshi A, Joshi KD, Sthapit BR. 1996. Farmer participatory crop improvement. I. Varietal selection and breeding methods and their impact on biodiversity. Experimental Agriculture **32**, 445–460.

Witcombe JR, Joshi KD, Gyawali S, Musa AM, Johansen C, Virk DS, Sthapit BR. 2005. Participatory plant breeding is better described as highly client-oriented plant breeding. I. Four indicators of client-orientation in plant breeding. Experimental Agriculture **41**, 299–319.

Witcombe JR, Petre R, Jones S, Joshi A. 1999. Farmer participatory crop improvement. IV. The spread and impact of a rice variety identified by participatory varietal selection. Experimental Agriculture **35**, 471–487.

Witcombe JR, Yadavendra JP. 2014. How much evidence is needed before client-oriented breeding (COB) is institutionalised? Evidence from rice and maize in India. Field Crops Research **167**, 143–152.

Worku M, De Groote H, Munyua B, *et al.* 2020. On-farm performance and farmers' participatory assessment of new stress-tolerant maize hybrids in Eastern Africa. Field Crops Research **246**, 107693.

Yami M, Cavicchioli M, Cole SM, Wossen T, Abdoulaye T. 2024. Why contexts matter for gender equal outcomes in research-based plant breeding: the case of maize in Nigeria. Economic Botany **78**, 445–464.

Zandalinas SI, Mittler R. 2022. Plant responses to multifactorial stress combination. New Phytologist **234**, 1161–1167.

Zeng D, Alwang J, Norton GW, Shiferaw B, Jaleta M, Yirga C. 2017. Agricultural technology adoption and child nutrition enhancement: improved maize varieties in rural Ethiopia. Agricultural Economics **48**, 573–586.

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